

JavaScript Functions - Simple Notes

1. Function Declaration

A function created using the `function` keyword.

```
function greet() {  
  console.log("Hello");  
}
```

✓ Can be called before declaration.

2. Function Expression

A function stored in a variable.

```
const greet = function() {  
  console.log("Hello");  
};
```

✗ Cannot be called before declaration.

3. Arrow Function

Short form of a function.

```
const greet = () => {  
  console.log("Hello");  
};
```

```
const add = (a, b) => a + b;
```

4. Parameters

Values received by a function.

```
function greet(name) {  
  console.log("Hello " + name);  
}
```

```
}
```

5. Return

Sends a value back from a function.

```
function add(a, b) {  
  return a + b;  
}
```

6. Scope

Scope = Where a variable can be used.

Global Scope

Accessible everywhere.

Local Scope

Accessible only inside the function.

Block Scope

Variables declared with `let` and `const` work only inside `{ }`.

7. Hoisting

JavaScript moves declarations to the top before execution.

```
hello();  
  
function hello() {  
  console.log("Hello");  
}
```

Function declarations are hoisted.

8. Closure

A function remembers variables from its outer function.

```
function outer() {  
  let count = 0;  
  
  return function() {  
    count++;  
    console.log(count);  
  };  
}
```

One-Line Revision

Topic	Meaning
Function Declaration	Normal function using function
Function Expression	Function stored in variable
Arrow Function	Short ES6 syntax
Parameter	Input to function
Return	Output from function
Scope	Variable accessibility
Hoisting	Declarations move to top
Closure	Remembers outer variables